Sequences — problem 9.54

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Prove that, for any positive integer $a_1 > 1$ there exists an increasing sequence of positive integers $a_1, a_2, a_3, \ldots$, such that $a_1^2 + \cdots + a_k^2$ is divisible by $a_1 + \cdots + a_k$ for all $k \geq 1$ (RO 1995).